The mobile revolution

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As an introduction to this themed edition on Mobile Systems, an overview is presented of the mobile radio technologies whose development and growth over the past decade have affected the lives of us all. The mobile phone has moved from the executive's trinket to become the essential tool of daily business life. It now sits on the threshold of becoming the mass-market means of personal communication for voice, data and video — all while the user is on the move world-wide.

1. Introduction

Mobile Systems, usually synonymous with the cellular telephone, have existed in the broadest sense since the earliest days of radio and the arrest of the murderer Crippen, the first criminal to be 'captured' by radio [1]. The 'Martini factor' — communications with anyone, anywhere and in any medium — has long attracted those who could afford a portable radio system, but until the past ten years the 'customers' have been primarily the military and the civil emergency services with some use of private mobile radio 'despatch' systems.

Two factors controlled the situation — the limited natural resource of radio spectrum and the technology to exploit it. The fundamental spectrum problem of how to provide sufficient capacity for a mass-market mobile telephone service was solved with the now classical publication in 1979 from Bell Laboratories [2] that proposed frequency reuse and the principal of an interference rather than noise-limited system design which opened the market for cellular radio. The technological challenge that remained was the exploitation of military communications technology at consumer electronics prices — we have only to see the products advertised in the newspapers and high street shops to realise the extent to which this has been met over the past few years.

Undoubtedly, mobility is one of the key themes for the future of telecommunications, it is an area where competition is well established and certain to increase, supported by an ever-increasing rate of technological change.

2. Vision, market and growth

There are a number of visions for the mobile radio future. The mobile vision, at least in the view of this author, is that of more mobile than fixed network connections and, while there is some way to go in realising this view, the point has already been reached where, on an annual basis, there are more new connections to mobile than to fixed networks.

The RACE Mobile programme, in 1990, produced a vision that within ten years 50% of calls would originate or terminate on a mobile handset — provided that the quality of service and price approached that of the fixed network. The third vision offered is that of the universal personal communicator:

- universal — services available everywhere,
- personal — pocketable terminal / tailored services,
- communicator — voice, data, paging, multimedia,

which also emanates from the early 1990s and pre-dates the recent explosive interest in the information superhighway. The ability to contact anyone, anywhere and at anytime has become a recognised customer need, as has the means to manage unwelcome calls. As we enter the 'information age', this customer need will include the ability to obtain any information, at any time, from anywhere, or indeed the ability to be alerted to information relevant to that customer.

The growth of cellular connections over the past decade, since its introduction in the UK has been spectacular as shown by the numbers in Fig 1.

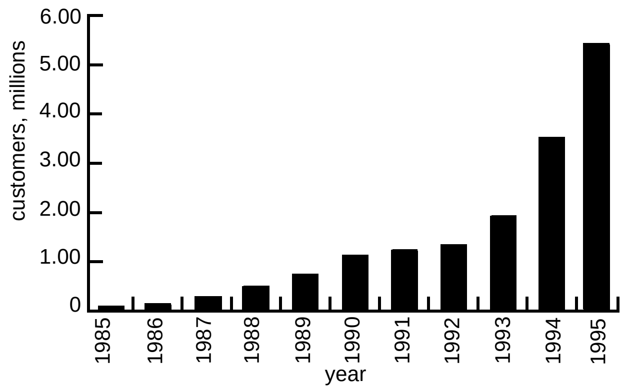

Fig 1 Growth of the UK cellular market.

As this paper is written the UK total has just exceeded 6 million connections to the now six analogue and digital UK networks (see Table 1).

Table 1 UK cellular networks.

	Analogue	Digital
Cellnet	TACS	GSM
Vodafone	TACS	GSM
One-2-One	-	DCS1800
Orange	-	DCS1800

These are made up from approximately four million analogue and two million digital connections. The penetration within the UK has now reached 10% of the population.

In terms of growth, fitting a simple exponential growth curve to the numbers of Fig 1 would predict that by the year 2001 the whole population of the UK would own at least one mobile phone! Current market estimates are in the region of 12—14 million and of course the market will saturate, but our marketing forebears failed miserably in their early predictions for saturation of the number of fixed telephone connections [3], so we have little reason to be cautious.

On a world scene, the total cellular connections now exceed 100 million and for GSM alone there are more than 20 million connections world-wide with predictions of over 100 million by the year 2000. In terms of market penetration there are some areas of the world — notably Scandinavia — where these are becoming very significant as shown in Fig 2.

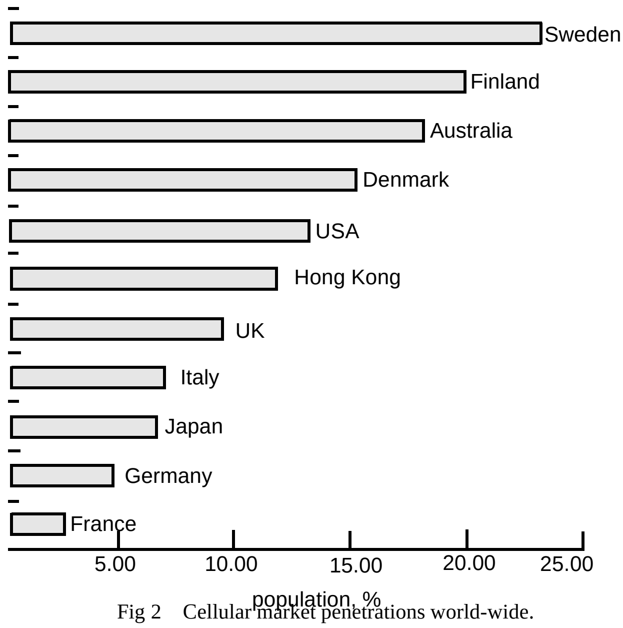

Fig 2 Cellular market penetrations world-wide.

All of which suggests that mobile is ‘big business’.

3. Radio technologies

Cellular is not the only radio technology of interest to mobile communications, nor does it address all the market segments. A range of technologies have been developed over the past 30 years or so, as indicated in Fig 3, which show two common features — all have or are emerging from analogue to digital system realisations and all expect to merge sometime early in the next century into third generation mobile systems (TGMS).

3.1 Radiopaging

Radiopaging is the most basic and well-established mobile communications service providing users with a range of options from simple tone to sophisticated alphanumeric pager units and it is interesting to note how the terminals are becoming fashion design accessories, including wrist watch variants. The market is still buoyant, particularly with the introduction of the non-rental service variants, and the move to digital European-wide paging is now under way using the ERMES [4] standard.

3.2 Cordless telephony

Cordless telephony became established with the analogue domestic units introduced in the mid-1980s and, indeed, is still well in evidence today; the UK standard was based on work carried out by BT in the early 1980s, being an adaptation of the US standard but with the addition of handset/base security code handshakes over the air path to

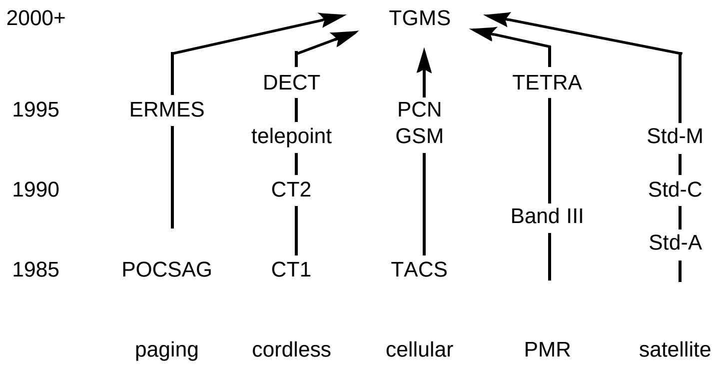

Fig 3 Radio technologies.

prevent improper access. Given the small fixed frequency allocation for this service, the density of handsets that can be supported, despite their low power levels, is limited, constraining a broader base of applications into business and public access (telepoint) operation. This limitation led to the development of the CT2 digital cordless standard supporting 40 channels in the frequency range 864—868 MHz. The CT2 standard was intended for domestic and wireless PBX applications and it was on this standard that the public access telepoint service was based.

3.3 Telepoint

The goal of telepoint was low-cost mobile telephony for the mass market and within the UK the government licensed four operating companies; however, within only a few years all had ceased trading, partly because the government had also licensed three PCN operating companies at much the same time. There were several reasons for the failure of telepoint within the UK:

- the service was predicated on an existing population of cordless (CT2) handsets, both domestic and business — in practice this population did not exist,
- the early technology did not operate to a common air interface so users could not roam from one system to another,
- there were late deliveries by the manufacturers,
- the service was too expensive, particularly given the limited number of base sites deployed and the one-way nature of the initial service,
- the UK culture is different from, say, Hong Kong where the service, initially at least, was a success.

The perceived competition from the emerging PCN networks also served to undermine the market. There was a window of opportunity which, had it been aggressively attacked could well have yielded the anticipated success.

3.4 DECT

The Digital European Cordless Telephony standard, now renamed Digitally Enhanced Cordless Telephony, followed close behind CT2. Its prime goal being to provide for the very high density of operation found in offices and to support higher bit rates for data applications. It uses a 12-channel TDMA digital radio system with a radio channel data rate of 1152 kbit/s on ten radio carriers operating over the range 1880—1900 MHz.

Applications for the digital cordless technologies include residential cordless handsets, wireless PBX, telepoint, cordless data, and local (wireless) access. One might conclude that CT2 is a maturing technology — residential products have existed for some time but are probably slightly too expensive as the market take-up has not been huge, despite their improved technical performance. The wireless PBX market for CT2 is beginning to expand. Residential DECT units are now emerging and the market response looks promising. DECT also offers a wider range of business applications with its ability to concatenate channels and support ISDN applications. The commercial case for telepoint was difficult to support against low-cost cellular, although the principle could still find application in third generation mobile systems.

Both digital cordless technologies are suited to wireless local loop applications in providing the ‘last 100 m’ of access to a fixed network — see Fig 4; however, while there have been a number of demonstrations and trials of such an approach, these have so far foundered on the regulatory rock. The topic is explored further in Merrett and Buttery [5] as is the ‘OnePhone’ concept, a combined DECT and GSM handset that offers the quality and tariffs of cordless

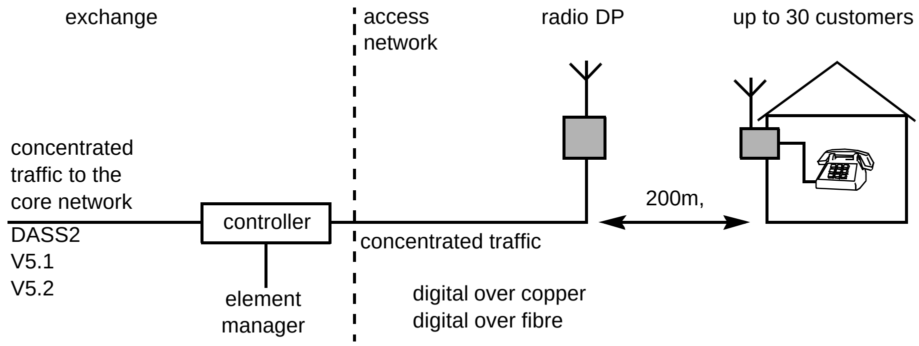

Fig 4 CT2 and DECT wireless local loop.

systems while in the office or at home, combined with the wide area flexibility of GSM cellular while on the move — a precursor of the third generation systems that are to come.

3.5 *Cellular telephony*

Cellular telephony was introduced in the UK in 1985 using an analogue standard known as TACS (Total Access Communications System) which was based on the US American Mobile Phone System (AMPS). Initially two companies, trading as Cellnet and Vodafone, were licensed to operate around 900 MHz and more recently two PCN operators, One-2-One and Orange were licensed to operate digital systems around 1800 MHz. Both Cellnet and Vodafone have also introduced digital systems based on the GSM standard, again operating around 900 MHz. The issues facing any operator at start up are those of radio and network planning to ensure coverage and as the network matures these issues migrate to those of capacity and infill - such issues are not trivial and the success of the operator depends very much on making the right choices in rolling out a network in a very dynamic market-place. Button et al [6] review these topics and outline some of the techniques adopted and the tools that have been developed within BT to address the issues.

3.6 *Global system for mobile (GSM)*

TACS is only one of a number of analogue cellular standards that exist within Europe and indeed world-wide and, as a result, handset roaming between countries is near non-existent, nor was there any evolutionary path to bring these systems to a common format. The desire for a common European standard for cellular telephony led to the development of GSM. Its prime goals were for mass-market use leading to low cost from the economy of scale of manufacture, roaming capability so that users were no longer tied to one network, digital speech with encryption, advanced fraud protection and a range of integrated services for voice, data and messaging. A key feature of GSM is that it defines a complete network — it is not just a radio interface as are the digital cordless technologies. The GSM network architecture is shown in Fig 5.

The success of GSM on a world-wide basis is a tribute to European collaboration, and developments are ongoing. There is still much activity as described in Holley [7]. Data developments are covered in Fenton et al [8] and the topic of speech and transmission requirements in cellular networks is addressed in Goetz [9]. GSM techniques have also extended into the airways with the adoption of the terrestrial flight telecommunications system now being deployed on European aircraft and described in Pettifor and Flanagan [10] along with their trans-oceanic satellite counterparts.

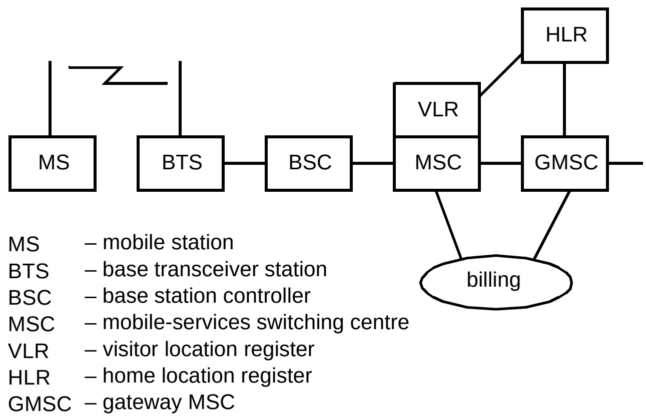

Fig 5 The GSM network architecture.

3.7 *Personal communications networks*

Personal communications networks (PCN) resulted from the DTI's document 'Phones on the Move' published in January 1989. Licence winners were announced at the end of that year and the initial three awards led to the launch of the One-2-One system in September 1993 and that of Orange in April 1994. The technology employed is that of GSM but frequency translated to operate at 1800 MHz and optimised for hand-portable terminals with power classes of

1W peak / 0.125W mean; this leads to smaller cell sizes, but given the objective of a high-capacity system for mass-market use this is not a disadvantage in any but rural areas.

3.8 Private mobile radio

Private mobile radio (PMR) does not normally form part of a public telecommunications network, although BT, like many other utilities, uses such systems for their internal operations with installation and maintenance people. Such use and its development is described in Garrett [11].

Existing PMR systems use analogue technology but are about to make the digital transition with the adoption of the European-wide TETRA standard — trans-European trunked radio. There is much interest within the public safety radio systems community for TETRA, particularly with the promise of its European-wide adoption. Within the larger picture PMR, too, is seen as being absorbed into TGMS.

3.9 Mobile satellite systems

Mobile satellite systems are seen as complementing terrestrial cellular and address two market segments. In the shorter term, before the advent of TGMS, they will provide the high-mobility customer and business traveller with a single handset that will operate world-wide providing global coverage with a single telephone number. They will also provide coverage both to territories with under-developed infrastructure and to those remote areas of the world where it will never prove economic to deploy conventional terrestrial cellular systems — the oceans, the polar regions and the deserts for example.

The need for integration of satellite into mobile communications can best be appreciated by taking an example such as Australia; its land mass is greater than that of Europe, it has a terrestrial cellular service that covers 85% of the population — yet this represents only 4.6% of the land mass!

Typical of mobile satellite systems, and one which has attracted much attention over the past few years is the Iridium System proposed by Motorola. It will employ some 66 satellites in low (780 km) polar orbital planes, each employing 48 spot beams and supporting voice communication at 4.8 kbit/s and data at 2.4 kbit/s. The key technical difference with such low-earth orbit (LEO) systems is that the user is, relatively, static compared with the velocity of the orbiting satellite. In effect, therefore, the mobile base station is moving across the user and as the area of each satellite's spot beam passes over the user, a handover must be initiated to the next satellite; this demands significant on-board processing and switching capability as well as direct satellite-to-satellite communication links. Mobile satellite systems and their broadband counterparts are discussed in Finean [12].

3.10 Third generation mobile systems

By the beginning of the next century, customers will be looking for an affordable, high-quality mobile communications system that provides both the services which are commonplace today and those that would be expected to be readily available in the year 2000, such as (mobile) videophone and high-speed data applications, as well as general-purpose telephony. Narrow and broadband ISDN-based services will have become both more available and will be widely used, and it is expected that customers will be seen managing and customising services to their own particular needs through the increased use of intelligent networks.

The future mobile communications user might thus expect to see:

- broadband and multimedia capability,
- flexible bandwidth-on-demand allocation,
- speech quality comparable with fixed network,
- packet data capabilities,
- global availability and global roaming,
- terrestrial and satellite capability in one system family,
- mass-market and niche-market products in one system.

The key differentiator in this, over second generation systems, is that of broadband service delivery, coupled with bandwidth-on-demand allocation, this latter, for example, being used to download a large file during a voice conversation or to secure (and pay for) a higher quality speech channel. Broadband is typically considered as an access rate of 2 Mbit/s, although it is recognised that for wide area mobile applications the rate may well fall to around 144 kbit/s.

4. Universality of service

The anticipated demand will be for a next-generation service which is universally available and encompasses both public and private systems. While it is likely that the air interface for TGMS will require a revolutionary development to support the broadband, bandwidth-on-demand service delivery, it is equally important that the supporting network infrastructure should be an evolutionary convergence of GSM, DECT, B-ISDN and IN.

It is expected that such systems will be progressively introduced during the early part of the next century and that, in time, the existing second generation systems will all migrate to TGMS as indicated in the Fig 6, which is taken from the recently published UMTS Task Force Report [13]. The general requirements for third generation mobile systems are discussed in Groves and Clapton [14] and a detailed review of the required network architecture is given in Cullen and Lobley [15].

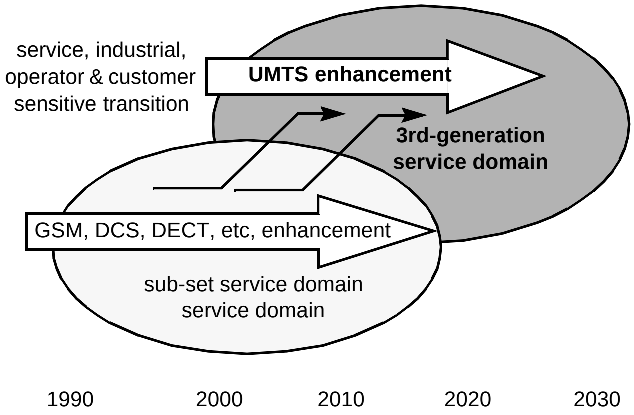

Fig 6 The move to third generation systems.

4.1 Radio spectrum

Radio spectrum is the key resource that enables any (and all) mobile communication system(s). It is a scarce resource that needs managing carefully and at a global level, particularly if objectives such as those of TGMS are to be realised. Hewitt [16] provides a detailed overview of radio spectrum and its management, as well as discussing the current issues surrounding spectrum allocations for mobile systems.

4.2 Environmental issues

The environmental issue complements the Hewitt paper [16] on spectrum and Tattersall [17] addresses difficult but topical issues of radio interference from mobile handsets, biological safety and the environmental impact of antennas and radio base stations.

5. Conclusions

Going back a little over ten years, a very different communications world from that of today can be seen:

- telex was used, not fascimile,
- people used typing pools rather than word processors,
- mainframes, not portable personal computers, were used,
- there was a limited market for consumer electronics,
- there was a low service culture,
- it was a protected telecommunications market,
- in the UK, there were only 10 000 car-based mobile phones.

Looking ten years out, a very different world can be envisaged:

- a very competitive telecommunications market,
- open standards for interconnect,
- a competitive, dynamic market for services,
- increased use of telecommunications in the home and at work,
- a huge increase in the use of mobile phones.

That the demand for mobile personal communications is ever increasing is not a difficult prediction, but how will it be met?

- in the view of this author at least, GSM/DCS will dominate cellular communications for the next ten years — certainly within Europe,
- the vision for the next generation of systems remains the universal personal communicator,
- TGMS will be achieved via a range of yet to be developed radio interfaces (including satellite) supported by a network that converges the technologies of GSM, broadband ISDN and the intelligent network,
- however, there is scope in the interim for hybrid second generation systems such as the GSM/DECT ‘OnePhone’ product.

Finally a challenge — the commercial success of any future mobile communications system will depend upon excellence of customer service — excellence in terms of coverage, of capacity and of quality — and all this cohered across a multimedia, multi-operator/multi-service provider global network.

That is the goal for the next decade.

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